

Celebal SQL Assignment 3

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Step 1: Database Setup

- Download the Superstore dataset from **KaggleHub** using Python.
- Save the dataset to the local working directory.
- Convert the dataset encoding to **UTF-8**.
- Import the dataset into the `superstore_raw` table in MySQL.
- Create the `customers`, `products`, and `orders` tables.

```
mysql> CREATE DATABASE superstore;
Query OK, 1 row affected (0.02 sec)

mysql> USE superstore;
Database changed
mysql> exit
Bye
manas@manas: ~$
```

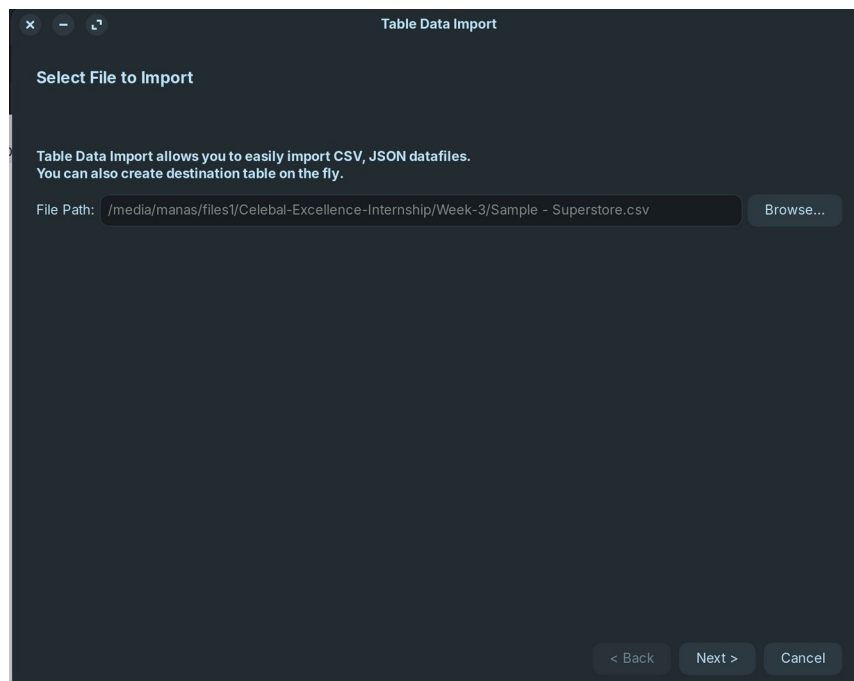


Table Data Import

Import Data

The following tasks will now be performed. Please monitor the execution.

Prepare Import

Import data file

Finished performing tasks. Click [Next >] to continue.

Message Log

- Data import

- Data import

- Data import

- Data import

- Data import

- Data import

- Data import

- Import finished

Import data file finished

Finished performing tasks.

Hide Logs

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Cancel

Table Data Import

Configure Import Settings

Detected file format: csv

Encoding: utf-8

Columns:

Source Column

Field Type

Row ID

int

Order ID

text

Order Date

text

Ship Date

text

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer I	Customer I	Segment	Country	City	State
1	CA-2016...	11/8/2016	11/11/2016	Second ...	CG-12520	Claire G...	Consumer	United S...	Henders...	Kentuc
2	CA-2016...	11/8/2016	11/11/2016	Second ...	CG-12520	Claire G...	Consumer	United S...	Henders...	Kentuc
3	CA-2016...	6/12/2016	6/16/2016	Second ...	DV-13045	Darrin V...	Corporate	United S...	Los Ang...	Californ

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Cancel

Step 2: Perform Required Queries

Question 1: Find all orders where sales are greater than the average sales (Subquery)

Query:

```
SELECT *  
FROM orders  
WHERE Sales > (  
    SELECT AVG(Sales)  
    FROM orders  
) ;
```

#	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Product ID	Sales	Quantity	Discount	Profit
1	4010	CA-2017-144463	1/1/2017	1/5/2017	Standard Class	SC-20725	FUR-FU-10001215	474.4300	11	0.0000	199.2606
2	766	CA-2014-162775	1/13/2014	1/15/2014	Second Class	CS-12250	TEC-AC-10003174	646.7400	6	0.0000	258.6960
3	768	CA-2014-162775	1/13/2014	1/15/2014	Second Class	CS-12250	OFF-ST-10000025	572.5800	6	0.0000	34.3548
4	2979	CA-2014-109232	1/13/2014	1/16/2014	Second Class	ND-18370	FUR-CH-10000422	545.9400	6	0.0000	87.3504
5	2880	CA-2016-152072	1/15/2016	1/19/2016	Standard Class	Dp-13240	TEC-AC-10000844	254.9700	3	0.0000	91.7892
6	876	CA-2016-116736	1/17/2016	1/21/2016	Standard Class	CC-12430	FUR-FU-10004017	322.5900	3	0.0000	64.5180
7	878	CA-2016-116736	1/17/2016	1/21/2016	Standard Class	CC-12430	TEC-AC-10002049	371.9700	3	0.0000	66.9546
8	2303	CA-2016-106558	1/17/2016	1/21/2016	Standard Class	DL-13495	TEC-AC-10001142	316.0000	4	0.0000	31.6000
9	1122	US-2014-147627	1/20/2014	1/26/2014	Standard Class	HL-15040	TEC-PH-10001061	699.9300	7	0.0000	181.9818
10	1127	US-2014-147627	1/20/2014	1/26/2014	Standard Class	HL-15040	FUR-CH-10002331	1067.9400	3	0.0000	224.2674
11	733	CA-2017-131954	1/21/2017	1/25/2017	Standard Class	DS-13030	OFF-ST-10000736	242.9400	3	0.0000	9.7176
12	2681	CA-2017-127026	1/21/2017	1/27/2017	Standard Class	MH-18115	TEC-AC-10002049	619.9500	5	0.0000	111.5910
13	516	CA-2017-127432	1/22/2017	1/27/2017	Standard Class	AD-10180	TEC-CO-10003236	2999.9500	5	0.0000	1379.9770
14	519	CA-2017-127432	1/22/2017	1/27/2017	Standard Class	AD-10180	OFF-ST-10004459	1126.0200	3	0.0000	56.3010
15	2580	US-2014-155502	1/26/2014	1/31/2014	Standard Class	SD-20485	OFF-ST-10004337	489.9200	8	0.0000	0.0000
16	1415	CA-2017-115994	1/28/2017	1/31/2017	Second Class	BT-11305	TEC-AC-10000580	239.9700	3	0.0000	26.3967
17	2290	CA-2017-115154	1/8/2017	1/11/2017	First Class	RS-19420	FUR-TA-10001950	892.9800	2	0.0000	80.3682
18	3209	CA-2016-108882	1/9/2016	1/15/2016	Standard Class	LA-16780	TEC-AC-10000420	349.9500	5	0.0000	118.9830
19	2216	CA-2017-129378	10/1/2017	10/2/2017	First Class	NS-18505	OFF-AP-10000576	241.9600	2	0.0000	60.4900
20	3296	CA-2017-113418	10/1/2017	10/3/2017	First Class	CM-12655	TEC-MA-10002428	1704.8900	11	0.0000	767.2005
21	4002	CA-2014-116834	10/11/2014	10/16/2014	Standard Class	Dp-13240	TEC-AC-10002942	345.0000	5	0.0000	58.6500
22	2917	CA-2015-134747	10/12/2015	10/17/2015	Second Class	DL-12925	TEC-PH-10001750	263.9600	4	0.0000	71.2692
23	2417	CA-2017-168655	10/12/2017	10/18/2017	Standard Class	ML-18040	TEC-AC-10002842	595.0000	5	0.0000	95.2000
24	811	CA-2014-151995	10/13/2014	10/15/2014	First Class	ZC-21910	FUR-TA-10002903	1298.5500	5	0.0000	311.6520

Question 2 : Find the highest sales order for each customer (Subquery)

Query:

```
SELECT o.*
FROM orders o
JOIN (
    SELECT
        `Customer ID`,
        MAX(Sales) AS MaxSales
    FROM orders
    GROUP BY `Customer ID`
) m
ON o.`Customer ID` = m.`Customer ID`
AND o.Sales = m.MaxSales
ORDER BY o.Sales DESC;
```

#	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Product ID	Sales	Quantity	Discount	Profit
6	4278	US-2016-107440	4/16/2016	4/20/2016	Standard Class	BS-11365	TEC-MA-10001047	9099.9300	7	0.0000	2365.9818
7	8489	CA-2016-158841	2/2/2016	2/4/2016	Second Class	SE-20110	TEC-MA-10001127	8749.9500	5	0.0000	2799.9840
8	2506	CA-2014-143917	7/25/2014	7/27/2014	Second Class	KL-16645	OFF-SU-10000151	8187.6500	5	0.0000	327.5060
9	6627	CA-2014-145541	12/14/2014	12/21/2014	Standard Class	TB-21400	TEC-MA-10001127	6999.9600	4	0.0000	2239.9872
10	510	CA-2015-145352	3/16/2015	3/22/2015	Standard Class	CM-12385	OFF-BI-10003527	6354.9500	5	0.0000	3177.4750
11	6521	CA-2017-138289	1/16/2017	1/18/2017	Second Class	AR-10540	OFF-BI-10004995	5443.9600	4	0.0000	2504.2216
12	7667	US-2016-140158	10/4/2016	10/8/2016	Standard Class	DR-12940	TEC-CO-10001449	5399.9100	9	0.0000	2591.9568
13	6341	CA-2017-143112	10/5/2017	10/9/2017	Standard Class	TS-21370	TEC-MA-10001047	5199.9600	4	0.0000	1351.9896
14	5885	CA-2016-136301	3/13/2016	3/15/2016	Second Class	EH-13765	OFF-SU-10000151	4912.5900	3	0.0000	196.5036
15	8991	US-2015-128587	12/24/2015	12/30/2015	Standard Class	HM-14860	TEC-CO-10003763	4899.9300	7	0.0000	2302.9671
16	1086	US-2016-143819	3/1/2016	3/5/2016	Standard Class	KD-16270	TEC-MA-10003979	4899.9300	7	0.0000	2400.9657
17	8205	CA-2015-114811	11/8/2015	11/8/2015	Same Day	KD-16495	TEC-MA-10000045	4643.8000	4	0.0000	2229.0240
18	2493	CA-2014-144624	11/19/2014	11/23/2014	Standard Class	JM-15865	TEC-PH-10002885	4548.8100	7	0.0000	1228.1787
19	9742	CA-2015-117086	11/8/2015	11/12/2015	Standard Class	QJ-19255	FUR-BO-10004834	4404.9000	5	0.0000	1013.1270
20	6621	US-2017-167402	1/13/2017	1/18/2017	Second Class	CP-12085	OFF-SU-10002881	4164.0500	5	0.0000	83.2810
21	3281	CA-2014-102988	4/5/2014	4/9/2014	Second Class	GM-14695	OFF-SU-10002881	4164.0500	5	0.0000	83.2810
22	2183	CA-2016-128818	5/7/2016	5/11/2016	Standard Class	CJ-12010	TEC-MA-10000488	3999.9500	5	0.0000	1159.9855
23	319	CA-2014-164973	11/4/2014	11/9/2014	Standard Class	NM-18445	TEC-MA-10002927	3991.9800	2	0.0000	1995.9900
24	7684	CA-2015-120782	4/28/2015	5/1/2015	First Class	SD-20485	OFF-BI-10003527	3812.9700	3	0.0000	1906.4850
25	5171	CA-2016-122903	5/27/2016	5/29/2016	Second Class	LA-16780	FUR-CH-10002024	3504.9000	5	0.0000	700.9800
26	7584	CA-2014-163223	3/21/2014	3/25/2014	Standard Class	KH-16690	TEC-PH-10000730	3499.9300	7	0.0000	909.9818
27	7915	CA-2017-165323	6/17/2017	6/21/2017	Standard Class	SR-20740	TEC-MA-10003673	3404.5000	5	0.0000	1668.2050
28	9858	CA-2015-164301	3/26/2015	3/30/2015	Standard Class	EB-13840	FUR-TA-10001889	3393.6800	8	0.0000	610.8624
29	252	CA-2016-145625	9/11/2016	9/17/2016	Standard Class	KC-16540	TEC-AC-10003832	3347.3700	13	0.0000	636.0003

Question 3: Calculate total sales for each customer (CTE)

Query:

```
WITH CustomerSales AS (  
    SELECT  
        `Customer ID`,  
        SUM(Sales) AS TotalSales  
    FROM orders  
    GROUP BY `Customer ID`  
)  
SELECT *  
FROM CustomerSales  
ORDER BY TotalSales DESC;
```

#	Customer ID	TotalSales	
1	SM-20320	25043.0500	
2	TC-20980	19052.2180	
3	RB-19360	15117.3390	
4	TA-21385	14595.6200	
5	AB-10105	14473.5710	
6	KL-16645	14175.2290	
7	SC-20095	14142.3340	
8	HL-15040	12873.2980	
9	SE-20110	12209.4380	
10	CC-12370	12129.0720	
11	TS-21370	11891.7510	
12	GT-14710	11820.1200	
13	BM-11140	11789.6300	
14	SV-20365	11470.9500	
15	CJ-12010	11164.9740	
16	CL-12565	10880.5460	
17	ME-17320	10663.7280	
18	KF-16285	10604.2660	
19	BS-11365	10501.6530	
20	EH-13765	10310.8800	
21	JL-15835	9799.9230	
22	GT-14635	9351.2120	
23	HW-14935	9300.2540	
24	TB-21400	9133.9900	

Question 4: Find customers whose total sales are above average (CTE + Subquery)

Query:

```
WITH CustomerSales AS (  
    SELECT  
        `Customer ID`,  
        SUM(Sales) AS TotalSales  
    FROM orders  
    GROUP BY `Customer ID`  
)  
SELECT  
    c.`Customer Name`,  
    ROUND(cs.TotalSales, 2) AS TotalSales  
FROM CustomerSales cs  
JOIN customers c  
ON cs.`Customer ID` = c.`Customer ID`  
WHERE cs.TotalSales > (  
    SELECT AVG(TotalSales)  
    FROM CustomerSales  
)  
ORDER BY TotalSales DESC;
```

#	Customer Name	TotalSales
1	Sean Miller	25043.05
2	Tamara Chand	19052.22
3	Raymond Buch	15117.34
4	Tom Ashbrook	14595.62
5	Adrian Barton	14473.57
6	Ken Lonsdale	14175.23
7	Sanjit Chand	14142.33
8	Hunter Lopez	12873.30
9	Sanjit Engle	12209.44
10	Christopher Conant	12129.07
11	Todd Sumrall	11891.75
12	Greg Tran	11820.12
13	Becky Martin	11789.63
14	Seth Vernon	11470.95
15	Caroline Jumper	11164.97
16	Clay Ludtke	10880.55
17	Maria Etezadi	10663.73
18	Karen Ferguson	10604.27
19	Bill Shonely	10501.65
20	Edward Hooks	10310.88
21	John Lee	9799.92
22	Grant Thornton	9351.21
23	Helen Wasserman	9300.25
24	Tom Boeckenhauer	9133.99

Question 5 : Rank all customers based on total sales (Window Function)

Query:

```
WITH CustomerSales AS (  
    SELECT  
        `Customer ID`,  
        SUM(Sales) AS TotalSales  
    FROM orders  
    GROUP BY `Customer ID`  
)  
SELECT  
    c.`Customer Name`,  
    ROUND(cs.TotalSales, 2) AS TotalSales,  
    DENSE_RANK() OVER (  
        ORDER BY cs.TotalSales DESC  
    ) AS CustomerRank  
FROM CustomerSales cs  
JOIN customers c  
ON cs.`Customer ID` = c.`Customer ID`  
ORDER BY CustomerRank;
```

#	Customer Name	TotalSales	CustomerRank
1	Sean Miller	25043.05	1
2	Tamara Chand	19052.22	2
3	Raymond Buch	15117.34	3
4	Tom Ashbrook	14595.62	4
5	Adrian Barton	14473.57	5
6	Ken Lonsdale	14175.23	6
7	Sanjit Chand	14142.33	7
8	Hunter Lopez	12873.30	8
9	Sanjit Engle	12209.44	9
10	Christopher Conant	12129.07	10
11	Todd Sumrall	11891.75	11
12	Greg Tran	11820.12	12
13	Becky Martin	11789.63	13
14	Seth Vernon	11470.95	14
15	Caroline Jumper	11164.97	15
16	Clay Ludtke	10880.55	16
17	Maria Etezadi	10663.73	17
18	Karen Ferguson	10604.27	18
19	Bill Shonely	10501.65	19
20	Edward Hooks	10310.88	20
21	John Lee	9799.92	21
22	Grant Thornton	9351.21	22
23	Helen Wasserman	9300.25	23
24	Tom Boeckenhauer	9133.99	24

Question 6 : Assign row numbers to each order within a customer (Window Function + PARTITION BY)

Query:

```
SELECT
    `Customer ID`,
    `Order ID`,
    Sales,
    ROW_NUMBER() OVER (
        PARTITION BY `Customer ID`
        ORDER BY Sales DESC
    ) AS RowNumber
FROM orders;
```

#	Customer ID	Order ID	Sales	RowNumber
1	AA-10315	CA-2016-103982	3930.0720	1
2	AA-10315	CA-2014-128055	673.5680	2
3	AA-10315	CA-2016-103982	431.9760	3
4	AA-10315	CA-2017-147039	362.9400	4
5	AA-10315	CA-2014-128055	52.9800	5
6	AA-10315	CA-2016-103982	41.7200	6
7	AA-10315	CA-2015-121391	26.9600	7
8	AA-10315	CA-2014-138100	14.9400	8
9	AA-10315	CA-2014-138100	14.5600	9
10	AA-10315	CA-2017-147039	11.5400	10
11	AA-10315	CA-2016-103982	2.3040	11
12	AA-10375	CA-2016-131065	499.9800	1
13	AA-10375	CA-2015-140921	149.9700	2
14	AA-10375	CA-2017-100230	116.9800	3
15	AA-10375	CA-2015-114503	84.9600	4
16	AA-10375	US-2017-169488	39.9600	5
17	AA-10375	CA-2014-130729	34.2720	6
18	AA-10375	CA-2015-140921	28.4000	7
19	AA-10375	CA-2017-100230	17.9400	8
20	AA-10375	US-2017-169488	16.9000	9
21	AA-10375	CA-2016-126613	16.7680	10
22	AA-10375	CA-2014-158064	16.5200	11
23	AA-10375	CA-2017-100230	14.9520	12
24	AA-10375	CA-2016-131065	8.2600	13

Question 7 : Display top 3 customers based on total sales (Window Function)

Query:

```
WITH CustomerSales AS (  
    SELECT  
        `Customer ID`,  
        SUM(Sales) AS TotalSales  
    FROM orders  
    GROUP BY `Customer ID`  
) ,  
RankedCustomers AS (  
    SELECT  
        `Customer ID`,  
        TotalSales,  
        DENSE_RANK() OVER (  
            ORDER BY TotalSales DESC  
        ) AS CustomerRank  
    FROM CustomerSales  
)  
SELECT  
    c.`Customer Name`,  
    ROUND(r.TotalSales, 2) AS TotalSales,  
    r.CustomerRank  
FROM RankedCustomers r  
JOIN customers c  
ON r.`Customer ID` = c.`Customer ID`  
WHERE r.CustomerRank <= 3  
ORDER BY r.CustomerRank;
```

#	Customer Name	TotalSales	CustomerRank
1	Sean Miller	25043.05	1
2	Tamara Chand	19052.22	2
3	Raymond Buch	15117.34	3

Step 3: Final Combined Query (JOIN + CTE + Window Function)

```
WITH CustomerSales AS (  
    SELECT  
        `Customer ID`,  
        SUM(Sales) AS TotalSales  
    FROM orders  
    GROUP BY `Customer ID`  
) ,  
CustomerRank AS (  
    SELECT  
        `Customer ID`,  
        TotalSales,  
        DENSE_RANK() OVER (  
            ORDER BY TotalSales DESC  
        ) AS CustomerRank  
    FROM CustomerSales  
)  
SELECT  
    c.`Customer Name`,  
    ROUND(cr.TotalSales,2) AS TotalSales,  
    cr.CustomerRank  
FROM CustomerRank cr  
JOIN customers c  
ON cr.`Customer ID` = c.`Customer ID`  
ORDER BY cr.CustomerRank;
```

#	Customer Name	TotalSales	CustomerRank
1	Sean Miller	25043.05	1
2	Tamara Chand	19052.22	2
3	Raymond Buch	15117.34	3
4	Tom Ashbrook	14595.62	4
5	Adrian Barton	14473.57	5
6	Ken Lonsdale	14175.23	6
7	Sanjit Chand	14142.33	7
8	Hunter Lopez	12873.30	8
9	Sanjit Engle	12209.44	9
10	Christopher Conant	12129.07	10
11	Todd Sumrall	11891.75	11
12	Greg Tran	11820.12	12
13	Becky Martin	11789.63	13
14	Seth Vernon	11470.95	14
15	Caroline Jumper	11164.97	15
16	Clay Ludtke	10880.55	16
17	Maria Etezadi	10663.73	17
18	Karen Ferguson	10604.27	18
19	Bill Shonely	10501.65	19
20	Edward Hooks	10310.88	20
21	John Lee	9799.92	21
22	Grant Thornton	9351.21	22
23	Helen Wasserman	9300.25	23
24	Tom Boeckenhauer	9133.99	24

Mini Project: Customer Sales Insights

Question 1 :- Top 5 customers

Query:

```
WITH CustomerSales AS (  
    SELECT  
        `Customer ID`,  
        SUM(Sales) AS TotalSales  
    FROM orders  
    GROUP BY `Customer ID`  
)  
SELECT  
    c.`Customer Name`,  
    ROUND(cs.TotalSales,2) AS TotalSales  
FROM CustomerSales cs  
JOIN customers c  
ON cs.`Customer ID` = c.`Customer ID`  
ORDER BY TotalSales DESC  
LIMIT 5;
```

#	Customer Name	TotalSales
1	Sean Miller	25043.05
2	Tamara Chand	19052.22
3	Raymond Buch	15117.34
4	Tom Ashbrook	14595.62
5	Adrian Barton	14473.57

Question 2 : Bottom 5 customers

Query;

```
WITH CustomerSales AS (  
    SELECT  
        `Customer ID`,  
        SUM(Sales) AS TotalSales  
    FROM orders  
    GROUP BY `Customer ID`  
)  
SELECT  
    c.`Customer Name`,  
    ROUND(cs.TotalSales,2) AS TotalSales  
FROM CustomerSales cs  
JOIN customers c  
ON cs.`Customer ID` = c.`Customer ID`  
ORDER BY TotalSales  
LIMIT 5;
```

#	Customer Name	TotalSales
1	Thais Sissman	4.83
2	Lela Donovan	5.30
3	Carl Jackson	16.52
4	Mitch Gastineau	16.74
5	Roy Skaria	22.33

Question 3 : Customers who made only one order

Query:

```
SELECT
    c.`Customer ID`,
    c.`Customer Name`,
    c.Segment
FROM customers c
JOIN (
    SELECT
        `Customer ID`
    FROM orders
    GROUP BY `Customer ID`
    HAVING COUNT(DISTINCT `Order ID`) = 1
) o
ON c.`Customer ID` = o.`Customer ID`;
```

#	Customer ID	Customer Name	Segment	
1	AO-10810	Anthony O'Donnell	Corporate	
2	AR-10570	Anemone Ratner	Consumer	
3	CJ-11875	Carl Jackson	Corporate	
4	JC-15385	Jenna Caffey	Consumer	
5	JR-15700	Jocasta Rupert	Consumer	
6	LD-16855	Lela Donovan	Corporate	
7	MG-18205	Mitch Gastineau	Corporate	
8	PH-18790	Patricia Hirasaki	Home Office	
9	RE-19405	Ricardo Emerson	Consumer	
10	RM-19750	Roland Murray	Consumer	
11	SM-20905	Susan MacKendrick	Consumer	
12	TC-21145	Theresa Coyne	Corporate	

Question 4 : Customers with above-average sales

Query:

```
WITH CustomerSales AS (  
    SELECT  
        `Customer ID`,  
        SUM(Sales) AS TotalSales  
    FROM orders  
    GROUP BY `Customer ID`  
)  
SELECT  
    c.`Customer Name`,  
    ROUND(cs.TotalSales,2) AS TotalSales  
FROM CustomerSales cs  
JOIN customers c  
ON cs.`Customer ID` = c.`Customer ID`  
WHERE cs.TotalSales > (  
    SELECT AVG(TotalSales)  
    FROM CustomerSales  
)  
ORDER BY TotalSales DESC;
```

#	Customer Name	TotalSales
1	Sean Miller	25043.05
2	Tamara Chand	19052.22
3	Raymond Buch	15117.34
4	Tom Ashbrook	14595.62
5	Adrian Barton	14473.57
6	Ken Lonsdale	14175.23
7	Sanjit Chand	14142.33
8	Hunter Lopez	12873.30
9	Sanjit Engle	12209.44
10	Christopher Conant	12129.07
11	Todd Sumrall	11891.75
12	Greg Tran	11820.12
13	Becky Martin	11789.63
14	Seth Vernon	11470.95
15	Caroline Jumper	11164.97
16	Clay Ludtke	10880.55
17	Maria Etezadi	10663.73
18	Karen Ferguson	10604.27
19	Bill Shonely	10501.65
20	Edward Hooks	10310.88
21	John Lee	9799.92
22	Grant Thornton	9351.21
23	Helen Wasserman	9300.25
24	Tom Boeckenhauer	9133.99